Nanopore Sequencing Workshop

Data Science Core

Alaska INBRE NIH IDeA (P20GM103395)

Data Generation, Template DNA for Nanopore Sequencing

- ZymoBIOMICS Oral Microbiome Standard (D6332)
- ZymoBIOMICS Gut Microbiome Standard (D6331)
- ZymoBIOMICS Microbial Community DNA (D6306)
- ZymoBIOMICS HMW DNA (D6322)

ZymoBIOMICS Oral Microbiome Standard (D6332)

Mimic the human oral microbiome

- 12 bacterial strains (6 Gram-negative, 6 Gram-positive)
- Present in a staggered abundance
- Includes four *Streptococcus* species.

useful for:

1. Determining specificity of species detection at different abundances
2. Assessing the effect of enriching or depleting for individual bacteria

ZymoBIOMICS Gut Microbiome Standard (D6331)

Mimic a human gut microbiome

- 18 bacterial strains (7 Gram-negative, 7 Gram-positive, one variable),
  2 fungal strains (*Saccharomyces cerevisiae* and *Candida albicans*),
- 1 archaeal strain (*Methanobrevibacter smithii*)
  Present in a staggered abundance
- Include five *Escherichia coli* strains

useful for:

1. Determining replicability of sequencing process
2. Assessing specificity and resolution within and across taxonomic groups, particularly for closely related strains or large genomes.

ZymoBIOMICS Microbial Community DNA Standard (D6306)

- 8 bacterial strains (3 Gram-negative, 5 Gram-positive)
- 2 fungal strains (*Saccharomyces cerevisiae* and *Cryptococcus neoformans*).
- Present in roughly equal theoretical genomic DNA proportions (12% each for bacteria, 2% for each yeast).
- Not specifically high molecular weight (it's purified genomic DNA, but not >50kb fragments)

useful for: Contrasting with other templates

ZymoBIOMICS HMW DNA Standard (D6322)

- 7 bacterial strains (3 Gram-negative, 4 Gram-positive)
- 1 yeast (*Saccharomyces cerevisiae*)
- Present in roughly equal theoretical genomic DNA proportions (14% each for bacteria, 2% for yeast).
- High Molecular Weight (HMW) DNA (typically >50kb), optimal for.

useful for: evaluating the performance of long-read sequencing technologies

Data Generation, Nanopore Sequencing Library Preparation

1. **Rapid Ramparts** (for Rapid with Adaptive Sampling Enrich): **Rapid Sequencing Kit** (SQK-RAD114) with Adaptive Sampling Enrichment (target amplification), **ZymoBIOMICS Oral Microbiome Standard (D6332)**

2. **Depletion Dynamos** (for Rapid with Adaptive Sampling Deplete): **Rapid Sequencing Kit** (SQK-RAD114) with Adaptive Sampling Depletion (removal of single target), **ZymoBIOMICS Oral Microbiome Standard (D6332)**

3. **Barcode Breakers** (for Rapid Barcoding): **Rapid Barcoding Kit** (SQK-RBK114-24) with 8 replicate samples **ZymoBIOMICS Gut Microbiome Standard (D6331)**

4. **Ligation Legends** (for Ligation): **Ligation Sequencing Kit** (SQK-LSK114), **ZymoBIOMICS Microbial Community DNA (D6306)**

Data Generation Groups

Rapid Ramparts	Depletion Dynamos	Barcode Breakers	Ligation Legends
RAD114 + AS Enrich	RAD114 + AS Deplete	RBK114	LSK114
Devin	**Kirsten**	**Libby**	**Ági**
David	Ayobami	Kyle	Yonatan
Felix	Maura	Soren	Anna
Gabriele	Michelle	Jason	Nathaniel
Susannah		James	Brijen

Printed Protocols with Instructor